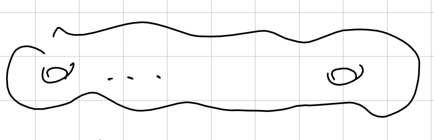
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



(n≥1)
↓
$\pi_1=0$.

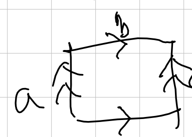
2-dim mfd

$L(p,q)$

$S^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

M ∨ N. ∧ ∨

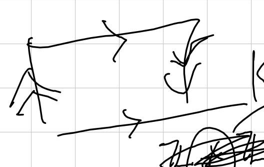
$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.



Klein. free product.

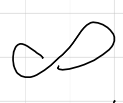
~~$\mathbb{Z}*\mathbb{Z}$~~

ababab b.

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

$ab=ba \rightarrow$ group ⟨⟩.



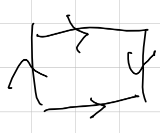
$\langle a,b \mid ab=ba\rangle$

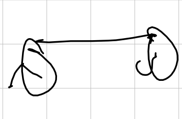
$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

$S^1\vee S^2$

$T: \mathbb{Z}\times\mathbb{Z}$





$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

$T\setminus\bullet$ $\mathbb{Z}*\mathbb{Z}$

$K\setminus\bullet$ $\mathbb{Z}*\mathbb{Z}$

theorem

S^1
↕
$\mathbb{R}$

$S^3\setminus\bullet : 0$.
↓
$\mathbb{R}^3 : 0$.

Theorem

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

M : n-dim
with (n≥3)

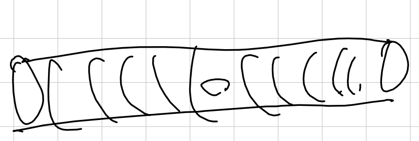
then $\pi_1(M) \simeq \pi_1(M\setminus\bullet)$.

$\simeq \pi_1(M\setminus 2pt)$

$\pi_1(S^n)=0$
$n\geq 2$.

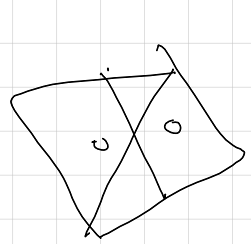
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$
$(n\geq 2)$.

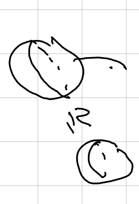
$\simeq \pi_1(M\setminus\{x_1, \ldots\})$.

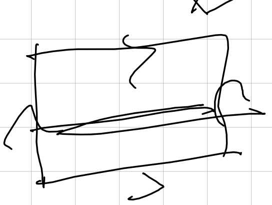


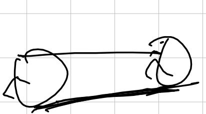
~~tree~~

$S^1\vee S^1\vee\mathbb{R}$.
↓
$\mathbb{Z}*\mathbb{Z}$



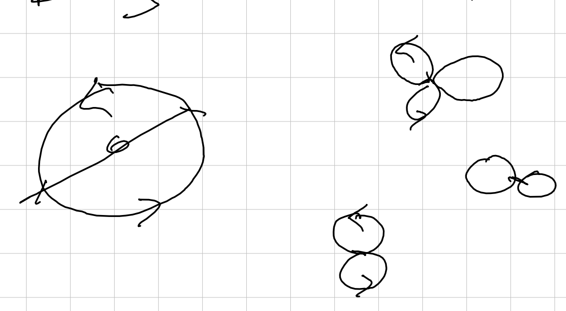




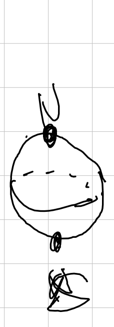


: $S^1\vee S^1\vee S^1$

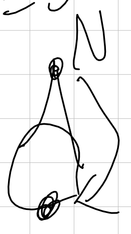
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

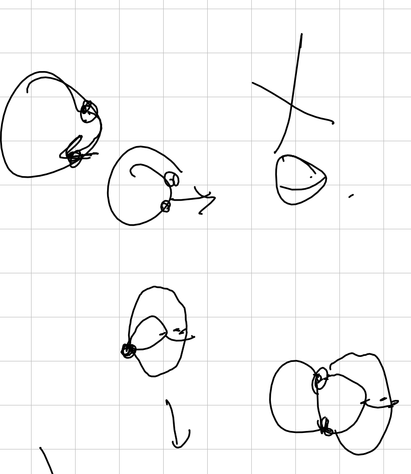


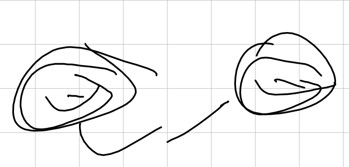
Example.



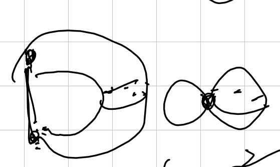
$S^2/N = S$



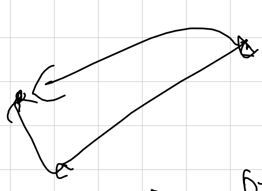


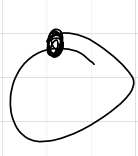


$\rightarrow \pi_1(\ldots)$



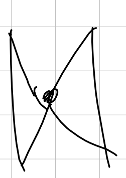
$S^1\vee S^2$,





S^1 : 0-cell 1개
1-cell 1개.

S^2 : 0-cell 1개
2-cell 1개



$S^1\vee S^1$

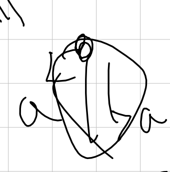
$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$\mathbb{Z}*\mathbb{Z}/a$

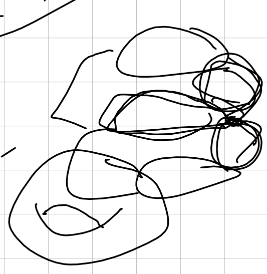
$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2\rangle$
$a^2=1$

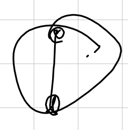


$\mathbb{RP}^\infty$: ~~0-cell 1개
1-cell 1개
2-cell 1개
3-cell 1개~~

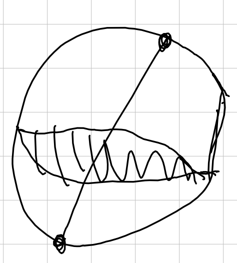
~~$\pi_1(\mathbb{RP}^\infty)$~~

$\pi_1(\mathbb{RP}^\infty)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

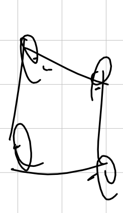


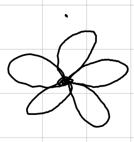


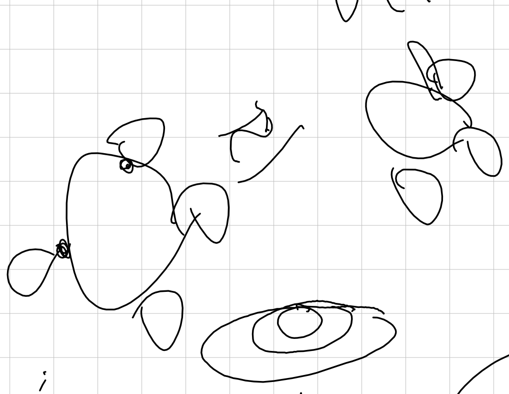
$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$



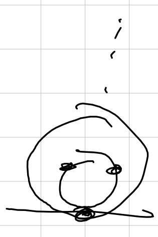
← $x^2+y^2+z^2=1$
∪ $x^2+y^2\leq 1$
∪ $(0,0,z)$ $-1\leq z\leq 1$.

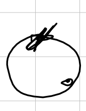


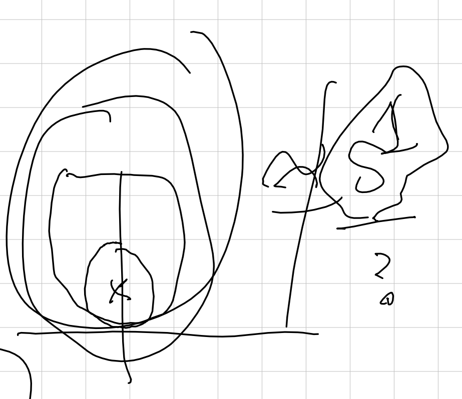


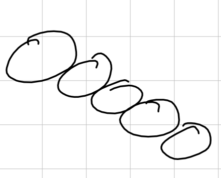


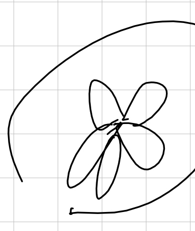
ℝ/ℤ

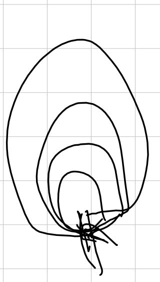


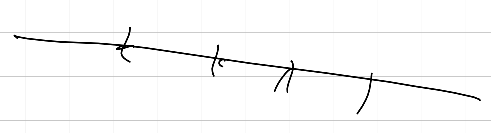






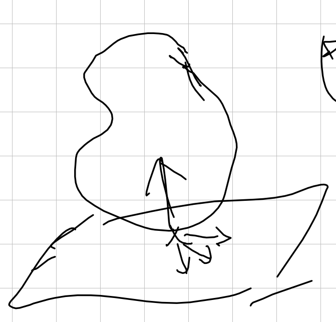




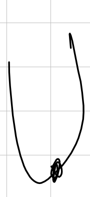


$z=x^4+y^4$
↓ exp

$K_1 = f(4,2)$ . .
$K_2 = f(4)$



$r=0$



$\frac{1}{x^4+z}$.

$\int \frac{R\,dz}{y}$

$\oint \frac{1}{z}\,dz$
$2\pi i$

$2\pi i \times (-1)$
‖
0